



Fig. 4: KICAD, a continuously-developed open source platform, can be used to create schematic diagrams and PCBs up to 32 copper layers (Image courtesy: www.kicad-pcb.org)

co-design and co-verification, concurrently designing and verifying hardware and software components of the system design, to deliver on more demanding time-to-market requirements, observes Vishal Abrol, senior group director, Field Engineering, Cadence Design Systems (India) Pvt Ltd. He says, "Concurrent design and verification allows for software development to begin before silicon is available, often before it is frozen, which can shave months off the software development schedule."

Launched in March 2015, Innovus Implementation System is for system companies that are working on complex, advanced node designs and face conflicting challenges around meeting requirements for power, performance and area (PPA) as well as turn-around time (TAT).

Abrol says, "It becomes very difficult to achieve the best power

performance with maximum productivity without compromising the outcome of either of these requirements. Innovus Implementation System provides a typical 10 to 20 per cent production-proven advantage in the PPA of the design, along with an up to 10 times TAT and capacity gain."

He adds, "The platform is the industry's first massively-parallel implementation solution. It is built on a next-generation platform with several integrations that result in an easy-to-use core implementation and sign-off flow that facilitates better engineering productivity."

Tools for board-level design

For every electronic product, a printed circuit board (PCB) is like its central nervous system, mechanically supporting it and electrically connecting electronic components mounted on the board. And, designers look to create a PCB that is competent in the first run. Therefore it is of paramount importance to design quality PCBs with the help of standard design tools in an efficient and cost-effective manner.

Tools in the market that look to tackle most of the complex design challenges and latest developments in some of these tools for board-level design are featured below:

OrCAD. In July 2014, Cadence introduced three new additions to its

OrCAD product line to accelerate the mainstream PCB design process and provide a significant boost to productivity and efficiency.

Abrol says, "The new OrCAD products include OrCAD Engineering Data Management (EDM), a comprehensive collaboration and management environment for OrCAD Capture; OrCAD Library Builder, a rapid automated part builder; and OrCAD Documentation Editor, an intelligent, automated PCB documentation environment."

PADS. Acquisition of PADS has strengthened Mentor Graphics' tools for board-level design. Recently in April, Mentor Graphics released three new tiers of the PADS offerings—PADS Standard, PADS Standard Plus and PADS Professional.

Standard and Standard Plus versions have the same graphical user interface (GUI) as the earlier versions to make it easier for the user to switch and adapt. These suites have been enhanced with additional capabilities to incorporate design archiving, common constraint management, integrated design and additional analysis capabilities.

But PADS Professional version has a completely different GUI as it is built on Xpedition design technology from Mentor Graphics. It is targeted at professionals who require more power and capabilities for highly-complex designs.

Later in the year, a 2D/3D co-design solution will be introduced along with other new features.

CADSTAR. The latest version of this single-board design tool from Zuken enhances design efficiency to meet the requirements of today's high-speed designs, as well as routing functionality and group handling with its new features. Items contained within a group can be selected and modified without first ungrouping these, which was not possible earlier. This simple improvement will provide users with more time to focus on their designs and improve design quality as ungrouping items

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